

BIS 629. Advanced Methods for Implementation and Prevention Science, Fall 2026

Tuesdays 3-4:50pm

Location: LEPH 101, 60 College Street

Instructor: Donna Spiegelman, Sc.D. Susan Dwight Bliss Professor of Biostatistics
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Department of Biostatistics
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Office Hours: Tuesdays 12P-2P at 135 College St, Suite 200, Room 228

Grader: Can Meng PhD Candidate
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Lecture: Classes meet once per week for one hour and fifty minutes.

Textbook: Course notes and readings,
Optional: Hayes, R. J., & Moulton, L. H. (2017). *Cluster randomised trials*;
Donner & Klar (2000). *Design and analysis of cluster randomization trials in health research*.
Casella and Berger (2001). *Statistical Inference(2nd). Chapter 8 and Chapter 10 (Chap 10.3)*.

Course Website: <http://canvas.yale.edu> – syllabus, lecture notes, computer code, readings, and homework assignments will be posted on the website.

Prerequisites: S&DS 5410 and S&DS 5420, or equivalent, or permission of the instructor. This course is intended for biostatistics graduate students in the second year and above and requires knowledge of, and comfort with, general mathematical statistics. Prior exposure to asymptotic theory, survival analysis, and/or longitudinal data analysis is desirable but not required. Basic statistical programming skills will also be helpful.

Course Description: The course presents methods for the design and analysis of studies arising in implementation and prevention science. These studies use a range of cluster-randomized designs, quasi-experimental designs, and observational designs. R will typically be used for course problem sets; students have the option of using SAS if they wish. The use of AI tools is encouraged. Students must submit the following when using AI: 1. State that it has been used, the name of the software and version; 2. In an Appendix, provide your prompts; 3. In the body of the homework, show how you verified the accuracy of any AI-generated information, reasoning, or output used in your homework. Students are responsible for the accuracy of all submitted work. Please note that AI makes many mistakes, and AI-generated content should not be accepted without independent verification. The commercial software

PASS is also used for many homework, and students will receive a free 6-month license for their use during the course.

Course Objectives: By the end of this course, students will be able to:

1. Understand the theory behind the major experimental and quasi-experimental designs used in implementation and prevention science
2. Apply this theory to produce sample size and power calculations for these designs
3. Apply this theory to analyze data from these designs

Grading:	% of overall grade
5 Homework Problem Sets	60%
Final exam	30%
Attendance	10%

Homework consists of a combination of theoretical development and problem sets using R, PASS, and AI tools. The theoretical portion will consist of mathematical derivation of methods. The practical portion requires summarizing important aspect of a data set that will be provided, along with the underlying R or PASS program, and/or AI prompts and responses. You should only include the relevant code, output and graphics in your homework. You may discuss problems with fellow students, but you must write up your own solutions, your own computer code and your own AI prompts.

Final exam is fully comprehensive of all material covered. It is a take-home, open-book exam.

Deadlines:

All assignments are due to the instructor through Canvas by the time stated on the syllabus. Exceptions to this policy may be granted if a deferral is formally requested in advance and documented by email. The instructor will grant deferrals sparingly and on a case-by-case basis for appropriately justified extenuating circumstances. Any assignments other than homework handed in after the due date/time will be given a 10% penalty for every 24 hours late in the absence of an approved deferral. Homework assignments submitted late will be penalized 1 point for each fraction of 24 hours after the due date, and assignments submitted after 48 hours will not be scored because the answer key will be provided at that time.

Course Schedule:

Randomized designs for Implementation and Prevention Science

9/8 Week 1: Introduction to implementation and prevention science; examples of implementation and prevention science studies for health promotion and clinical quality improvement; review of asymptotic hypothesis testing theory

Readings:

Brown, C. H., Curran, G., Palinkas, L. A., Aarons, G. A., Wells, K. B., Jones, Collins, L. ., Duan, N., Mittman, B. S., Wallace, A., Tabak, R. G., Ducharme, L., Chambers, D. A., Neta, G., Landsverk, J., Cheung, K., Cruden, G. (2017). An overview of research and evaluation designs for dissemination and implementation. *Annual review of public health*, 38, 1-22.

Casella and Berger. Statistical Inference. Pacific Grove, CA. Duxbury Press, 2001. Chapters 8 and 10.

9/15 Week 2 Review of asymptotic hypothesis testing theory, continued

Optional Readings:

Equivalence and Noninferiority Testing Using SAS/STAT® Software

<https://support.sas.com/resources/papers/proceedings15/SAS1911-2015.pdf>

Testing Hypotheses for Equivalence and Non-inferiority with Binary and Survival Outcomes

<https://www.sas.com/content/dam/SAS/support/en/sas-global-forum-proceedings/2020/4641-2020.pdf>

Liu, J. P., Hsueh, H. M., Hsieh, E., & Chen, J. J. (2002). Tests for equivalence or non-inferiority for paired binary data. *Statistics in medicine*, 21(2), 231-245.

Homework 1 distributed; due on 9/29.

9/22 Week 3: Theory for sample size and power calculations for cluster randomized trials

Readings:

Self, Steven G., and Robert H. Mauritsen. "Power/Sample Size Calculations for Generalized Linear Models." *Biometrics*, vol. 44, no. 1, 1988, pp. 79–86. JSTOR, [jstor.org/stable/2531897](https://www.jstor.org/stable/2531897). Accessed 6 Sept. 2021.

Steven G., Robert H. Mauritsen, and Jill Ohara. "Power calculations for likelihood ratio tests in generalized linear models." *Biometrics* (1992): 31-39.

9/29 Week 4: Randomization and causal inference in implementation research

Required readings:

Greenland, Sander, and James Robins. "Invited commentary: ecologic studies—biases, misconceptions, and counterexamples." *American journal of epidemiology* (1994): 747-760.

Detry, Mchelle A., and Roger J. Lewis. "The intention-to-treat principle: how to assess the true effect of choosing a medical treatment." *JAMA* (2014): 85-86.

Hernán, Miguel A., and Sonia Hernández-Díaz. "Beyond the intention-to-treat in comparative effectiveness research." *Clinical trials*1 (2012): 48-55.

Hernán, Miguel A., and James M. Robins. "Per-protocol analyses of pragmatic trials." *N Engl J Med*14 (2017): 1391-1398.

Homework 1 due start of class.

10/6 Week 5: Cluster-randomized trials (continuous, binary, varying cluster sizes, matching, stratification, impact of ICC, repeated measures, software)

Homework 2 distributed; due on 10/20.

Required readings:

Ritz J, Spiegelman D. Equivalence of conditional and marginal regression models for clustered and longitudinal data[J]. *Statistical Methods in Medical Research*, 2004, 13(4): 309-323.

Rutterford, C., Copas, A., & Eldridge, S. (2015). Methods for sample size determination in cluster randomized trials. *International journal of epidemiology*, 44(3), 1051-1067.

Raab, G. M., & Butcher, I. (2001). Balance in cluster randomized trials. *Statistics in medicine*, 20(3), 351-365.

Shih, W.J. (1997), Sample Size and Power Calculations for Periodontal and Other Studies with Clustered Samples Using the Method of Generalized Estimating Equations. *Biom. J.*, 39: 899-908. <https://doi.org/10.1002/bimj.4710390803>.

10/13 Week 6: Cluster-randomized trials continued

-- Mid-semester feedback form due

10/20 Week 7: Cluster-randomized trials continued; Constrained Randomization in Cluster Randomized Trials, Guest lecture, Dr. Fan Li, Associate Professor, Biostatistics, YSPH

Homework 2 due start of class.

Homework 3 distributed; Due on 11/3

Required reading:

Moulton, L. H. (2004). Covariate-based constrained randomization of group-randomized trials. *Clinical Trials*, 1(3), 297-305.

Li, F., Lokhnygina, Y., Murray, D. M., Heagerty, P. J., & DeLong, E. R. (2016). An evaluation of constrained randomization for the design and analysis of group-randomized trials. *Statistics in medicine*, 35(10), 1565-1579.

Optional readings:

Li, F., Turner, E. L., Heagerty, P. J., Murray, D. M., Vollmer, W. M., & DeLong, E. R. (2017). An evaluation of constrained randomization for the design and analysis of group-randomized trials with binary outcomes. *Statistics in medicine*, 36(24), 3791-3806.

Greene, E. J. (2017). A SAS macro for covariate-constrained randomization of general cluster-randomized and unstratified designs. *Journal of statistical software*, 77(CS1).

Yu, H., Li, F., Gallis, J. A., & Turner, E. L. (2019). cvcrand: A Package for Covariate-constrained Randomization and the Clustered Permutation Test for Cluster Randomized Trials. *R Journal*, 9(2).

Gallis, J. A., Li, F., Yu, H., & Turner, E. L. (2018). cvcrand and cptest: Commands for efficient design and analysis of cluster randomized trials using constrained randomization and permutation tests. *The Stata Journal*, 18(2), 357-378.

10/27 Week 8: Factorial and fractional factorial designs; MOST; SMART; Guest lecture, Dr. Raul Hernandez-Ramirez, Research Scientist, Department of Biostatistics, YSPH

Required readings:

Guastaferro, K., & Collins, L. M. (2019). Achieving the goals of translational science in public health intervention research: the Multiphase Optimization Strategy (MOST). *American Journal of Public Health*: S128-S129

Collins, L. M. (2018). Introduction to the factorial optimization trial. In *Optimization of behavioral, biobehavioral, and biomedical interventions: The Multiphase Optimization Strategy (MOST)* (Chapter 1, pp. 1-34). Springer, Cham.

Collins, L. M. (2018). Conceptual introduction to the multiphase optimization strategy (MOST). In *Optimization of behavioral, biobehavioral, and biomedical interventions: The Multiphase Optimization Strategy (MOST)* (Chapter 3, pp. 67-113). Springer, Cham.

11/3 Week 9: Stepped wedge designs (continuous, binary, cohort)

Homework 3 due start of class.

Homework 4 distributed; due on 11/17.

Readings:

Zhou X, Liao X, Kunz LM, Normand ST, Wang M, Spiegelman D. A maximum likelihood approach to power calculations for stepped wedge designs of binary outcomes. *Biostatistics*. 2018.

Hussey, M. A. and J. P. Hughes (2007). "Design and analysis of stepped wedge cluster randomized trials." *Contemp Clin Trials* 28(2): 182-191.

Li, F., Hughes, J. P., Hemming, K., Taljaard, M., Melnick, E. R., & Heagerty, P. J. (2021). Mixed-effects models for the design and analysis of stepped wedge cluster randomized trials: An overview. *Statistical Methods in Medical Research*, 30(2), 612-639.

swdpwr: A SAS Macro and R Package for Power Calculations in Stepped Wedge Cluster Randomized Trials. Chen J, Zhou X, Li F, Spiegelman *Comput Methods Programs Biomed.* 2022 Jan;213:106522

11/10 Week 10: Stepped wedge designs, continued

11/17 Week 11: LAGO (theory, example, design) – Guest lecture, Dr. Jingyu Cui, Post-doctoral Fellow, Department of Biostatistics and Center on Methods for Implementation and Prevention Science, Yale School of Public Health, jingyu.cui@yale.edu

Homework 4 due start of class.

Homework 5 distributed; due on 12/8.

12/1 Week 12: Hybrid designs

1. Curran, G. M., Bauer, M., Mittman, B., Pyne, J. M., & Stetler, C. (2012). "Effectiveness-implementation hybrid designs: combining elements of clinical effectiveness and implementation research to enhance public health impact". *Medical care*, 50(3), 217
2. Owen MA, Curran GM, Smith JD, Tedla Y, Cheng C, Spiegelman D. Power and Sample Size Calculations for Cluster Randomized Hybrid Type 2 Effectiveness-Implementation Studies. *Stat Med.* 2025 Feb 28;44(5):e70015. doi: 10.1002/sim.70015. PMID: 39930740.<https://onlinelibrary.wiley.com/doi/10.1002/sim.70015>.

Required readings:

Nevo, D., Lok, J. J., & Spiegelman, D. (2021). Analysis of “learn-as-you-go”(LAGO) studies. *The Annals of Statistics*, 49(2), 793-819.

Spiegelman, D., Nevo, D., Chakrabarti, A., Barnhart, D., Ganguli, B., & Lok, J. J. "Evaluating Public Health Interventions 10: Let's learn as we go, and minimize large scale public health intervention failures: the LAGO Design for optimizing complex multi-component interventions." (*Unpublished manuscript*).

12/8 Week 13: Discrete Choice Experiments, Guest lecture, Dr. Drew Cameron, Department of Health Policy and Management, YSPH

Homework 5 due start of class.

1. Street, D. J., et al. (2005). "Quick and easy choice sets: Constructing optimal and nearly optimal stated choice experiments." *International Journal of Research in Marketing* 22(4): 459-470.
2. Cameron DB, Ray I, Parida M, Dow WH (2023) Product preferences and willingness to pay for potable water delivery: Experimental evidence from rural Bihar, India. *PLoS ONE* 18(4): e0283892.
<https://doi.org/10.1371/journal.pone.0283892>

Final examination distributed

Final examination due on Tuesday, December 22, at 5P.

Honor Code: All students are expected to complete their own assignments and exams. Any student suspected of violating the code of academic and professional integrity will be **IMMEDIATELY** referred to the Committee on Academic and Professional Integrity. Please refer to the following website for behaviors that are subject to disciplinary action

<http://bulletin.printer.yale.edu/htmlfiles/publichealth/academic-policies.html>